

HAT-P-36b

R-0842

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_241126 · created 2026-07-28T19:23:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460640.9553 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.029
Transit depth (Rp/Rs)^2	2.47 +/- 0.91 [%]
Orbital Inclination (inc)	89.7 +/- 2.43
Airmass coefficient 1 (a1)	1.428 +/- 0.02
Airmass coefficient 2 (a2)	-0.1167 +/- 0.0057
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	None
Optimal Aperture	3.83
Optimal Annulus	5.94
Transit Duration (day)	0.0908 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.029 vs 0.126 (deviation 1.07σ)
Duration fit vs published	0.0908 d vs 0.0925 d (deviation 0.37σ)
Mid-time O–C	-1.3 min
Depth SNR	5.4
Residual scatter	1.38 %

Light curve

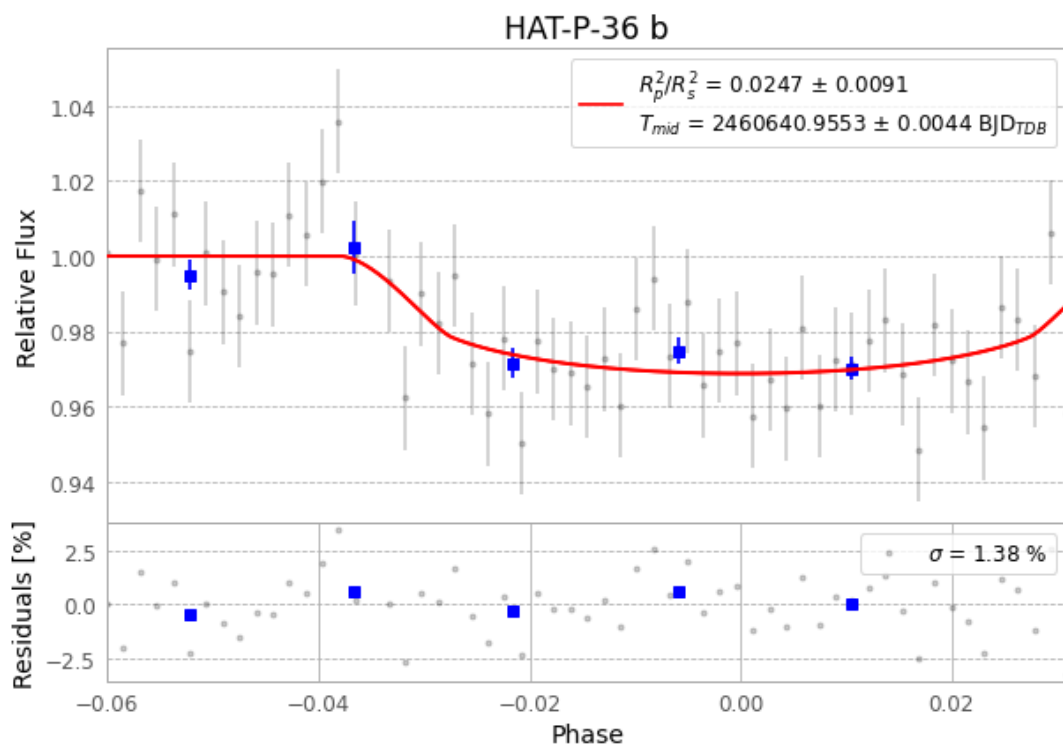


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-11-26.png (SHA-256 3c8b90fe54551f9a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 159], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df9c7b5535b597ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

59 FITS frames (39 MB), HATP-36241126090015.FITS → HATP-36241126115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-241126.log (e2deebd3e10c...), FinalLightCurve_HAT-P-36 b_2024-11-26.png (3c8b90fe5455...), FinalLightCurve_HAT-P-36 b_2024-11-26.csv (457bdb648dd2...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 19:23Z.